Seyun Jeong
Dr. Stylianou
4 May 2026

Final Report

Collection Strategy:

I used YouTube as my main source because it had many public hotel-room tours, reviews, guest vlogs, and travel videos. I considered Instagram and TikTok, but I decided to stay with YouTube because YouTube already had many hotel-room videos from different types of uploaders, including room reviewers, travel vloggers, hotel channels, and guests. My search terms included “hotel room interior”, “hotel room tour”, and “hotel room suite review.” I also repeated these searches with country names. I also filtered by recent uploads so that I was not only collecting the most popular or most recommended videos.

Challenges:

I first considered CSV, but switched to JSONL because the metadata was not flat. Each video can have multiple interior intervals, and each discriminative element has an object label, description, timestamp range, and occlusion flag. JSONL was still readable, but it was easier to process with Python. Gemini could not reliably decide which segments should count as hotel-room interiors without human guidance. Therefore, I provided specific timestamp ranges to Gemini so that it could analyze the scenes better. Upon reviewing the draft of the contents of the schema from Gemini, I validated each video afterward to ensure accuracy and consistency of the outputs.

Key Decisions:

A video was good enough to include if it had at least 10 seconds of usable hotel-room interior footage, and more importantly, I also considered whether the footage gave enough visual detail of the interior. I looked for visible room details such as beds, headboards, desks, bathroom fixtures, and general room layout because these details help describe the room. I also needed to identify evidence so that I could trust the hotel label. This could include the hotel name on signage, a TV welcome screen, branded objects, room materials, captions, or the YouTuber saying the hotel name. For example, in video_16, I used a printed card with the Waldorf Astoria logo as supporting identifying evidence, instead of relying only on the YouTube title. This mattered because the dataset should not just collect videos that look like hotels; it should collect videos that have enough visual and identifiable evidence for later frame extraction and analysis.

I avoided videos that were slideshows and had less than 10 seconds of usable room footage. For example, I excluded a slideshow-based video (Example1), a video with less than 10 seconds of usable room detail (Example2), and a video with limited visible hotel-room objects because it did not show the bathroom (Example3). I also excluded the interior of the room if it shows part of the lobby, hallway, exterior, balcony, or outside-only view. For people in the frame, I did not reject every video automatically because the room was still visible enough, and later occlusion is needed for the sake of privacy. Instead, I kept a video only when the room was still visible enough, and I recorded person_in_frame_seconds in the metadata if the person was visible. I recorded people because they may need to be occluded for privacy and can distract from hotel-identifying features.

One judgment call was deciding what counted as a hotel-room interior.
I included bedroom, bathroom, shower, sink area, closet, small kitchen area, and in-room entry area. I excluded lobby, hallway, exterior, balcony-only footage, outside-only window view, cover page, static slide, and end screen. One of the rules that I came up with for the room interior view is that even if a person stands inside the room showing the outside of the window, I counted it as part of the interior because outside of the window could be visible inside the hotel; however, I decided to exclude it because I focused more on what's visible inside the interior.

The most important fields were identity_evidence, interior_intervals, visible_objects, discriminative_elements, person_in_frame_seconds, and caveats. identity_evidence explains why I trust the hotel label. interior_intervals mark usable room footage. discriminative_elements record visually distinctive room details with timestamp ranges. The metadata was not only descriptive. It directly supported Part 2. interior_intervals controlled which parts of the videos were used for frame extraction, and discriminative_elements became a proxy for evaluating whether selected keyframes captured important room details.

Dr. Stylianou's "Hotels-50K" paper shows that hotel recognition faces high intraclass variation (rooms within the same hotel can look very different due to renovations or amenities) and low interclass variation (rooms from different hotels in the same chain can look very similar) (729). Therefore, discriminative elements matter in this assignment: they capture the visually distinctive features that distinguish one specific hotel from another, even when broader room features are shared across a chain.

Limitations:

YouTube gives me variety across channels and filming styles, although I still treat the YouTube-only sources as a dataset limitation. The dataset is limited because it is based on YouTube videos. This gives variety across uploaders, but it is still only one platform. The dataset is biased toward videos that travelers, reviewers, hotels, or vloggers choose to publish. These videos may have better lighting, more intentional framing, and more complete room views than real investigative footage. The dataset is also not globally balanced. Some countries, channels, hotel types, and creator styles are represented more than others (e.g., video_17, 18, 19 are all from one Vietnam-focused channel). Some videos include people, mirror reflections, personal belongings, room numbers, or guest names (e.g., video_03, video_04 capture room numbers verbally and on plaques). I recorded these matters in caveats when I saw them. The metadata was created by Gemini first and then validated by me. I manually checked interior_intervals thoroughly for videos 1-14 and the 5 videos used in Part 2 keyframe extraction, and less thoroughly for the rest. person_in_frame_seconds and discriminative_elements with their timestamps were drafted by Gemini, which I spot-checked but did not validate exhaustively. As a result, these fields involve approximations and subjective judgment calls, yet this hybrid approach also captured nuance that I could have missed. Collaboration between Gemini drafting and human validation could be a powerful approach for further research, giving both a nuanced description and efficiency. Some interior_intervals may include brief window-outside views that I have not yet audited.

Analysis and reflection:

My approach was uniform temporal sampling with 10 keyframes per video. Frames were extracted at 1 fps (per second) from each video's interior_intervals, then 10 evenly spaced frames were selected from that pool. I chose this method to keep the pipeline simple, reproducible, and focused on building a robust baseline first. The assignment itself preferred a well-motivated, simple method over a complex one I did not fully understand, so I chose one I could work from now and expand later.

videos	metric numbers	visual numbers
video_09	4/5	5/5
video_11	2/5	4/5
video_12	2/5	2/5
video_21	0/5	1/5
video_29	2/5	5/5

total	10/25	17/25

The pipeline produced 50 keyframes total (5 videos × 10). The table above demonstrates that metric and visual numbers diverge for most videos. Even where counts match (video_12: 2/5 in both), the specific elements captured differ. Across all five videos, the metric scored 10/25, while visual inspection found 17/25, which is a 28 percentage-point gap. This means visual inspection captured more than the timestamp-overlap metric credits. Overall, the keyframes do capture the essential aspects of most rooms, but they miss features that appear only in narrow time windows. video_21 is the weakest example, with only the sink visually identifiable.

Strengths:

First, the method is simple, reproducible, and does not require a GPU. Second, it achieves strong compression; 93 percent reduction with only 7 percent of frames kept. Third, the metric/visual gap finding itself is a strength of the evaluation process: it shows what the metric misses and helps validate the metric's accuracy.

Weaknesses:

First, the method has no content awareness; it picks frames by time, not by what is visible. Second, the coverage metric understates real visual coverage, with a 28 percentage-point gap between the metric (10/25) and visual inspection (17/25). Third, some annotation timestamps in the metadata are imprecise (noticed during inspection of video_12 and video_29), which compounds the metric's strictness.

Failure case:

video_21 is the weakest example, with only 1/5 elements visually identifiable. Narrow-window features such as the LG Styler, washing machine, bidet, and Eco Nature dispensers all fell outside selected timestamps. Redundancy is not a major issue at this scale, since each video produced 126–170 extracted frames at 1 fps, and we selected only 10 keyframes from that pool. If I extended this work, I would explore the other approaches the assignment suggests: frame-difference or motion-based selection, clustering-based approaches, embedding-based selection, and shot boundary detection. These could address the narrow-time-window misses that uniform sampling cannot catch.

Efficiency:

Through validation, I found that keyframe selection took around 0.05 seconds via select_keyframes.py and evaluate.py took approximately 0.02 seconds, both well under a second. In my cached local run, download.py took 6.27 seconds because the videos were already present locally, while extract_frames.py took 46.67 seconds and was the slowest measured processing step. Keyframe selection is fast because it just reads timestamps, sorts them, computes indices, and copies files; it does not run AI or computer vision models on the frames. The main scaling costs are video downloading and frame decoding, which grow with the number and length of videos. Selection itself should scale well; extraction and downloading will scale less easily, especially with longer videos.

Acknowledgments:

I used AI tools as support during this project, while making the final dataset, validation, and analysis decisions myself. I used ChatGPT and Claude for organizing my thoughts and asking process questions, such as "CSV looks not organized, and there are multiple interior_intervals that I want to implement, what will be the best solution?" I used Gemini to draft initial metadata entries for each video after realizing that watching every video in full detail took me a lot of time, then manually reviewed and corrected the outputs, especially the interior intervals and the five videos used for keyframe extraction. Future work might utilize the Google Gemini API to process multiple videos at the same time. I also used Cursor's embedded AI agent to reformat metadata.jsonl into a more human-readable layout so I could review and modify entries when I found issues. For technical background on keyframe extraction, I searched "key frame extraction video" on Google and learned about FFmpeg from introductory YouTube tutorials. English is my third language, so I received writing consultation from Saint Louis University's INTO English Program Writing Service (consultant: Everett Wilhelm). The final schema decisions, inclusion/exclusion decisions, keyframe inspection, and interpretation of results are my own.

Works Cited

Stylianou, Abby, et al. "Hotels-50K: A Global Hotel Recognition Dataset." *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, no. 1, 2019, pp. 726-33, https://doi.org/10.1609/aaai.v33i01.3301726.

FFmpeg. FFmpeg Project, ffmpeg.org. Accessed 3 May 2026.

yt-dlp. yt-dlp Project, github.com/yt-dlp/yt-dlp. Accessed 3 May 2026.